

Online Library System - ER Diagram

ER Diagram for an Online Library System

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Objectives

- • Understand ER modeling
- • Identify entities, attributes, PK & FK
- • Design Online Library database
- • Explain relationships clearly

Member

Book

Author

Main Entities

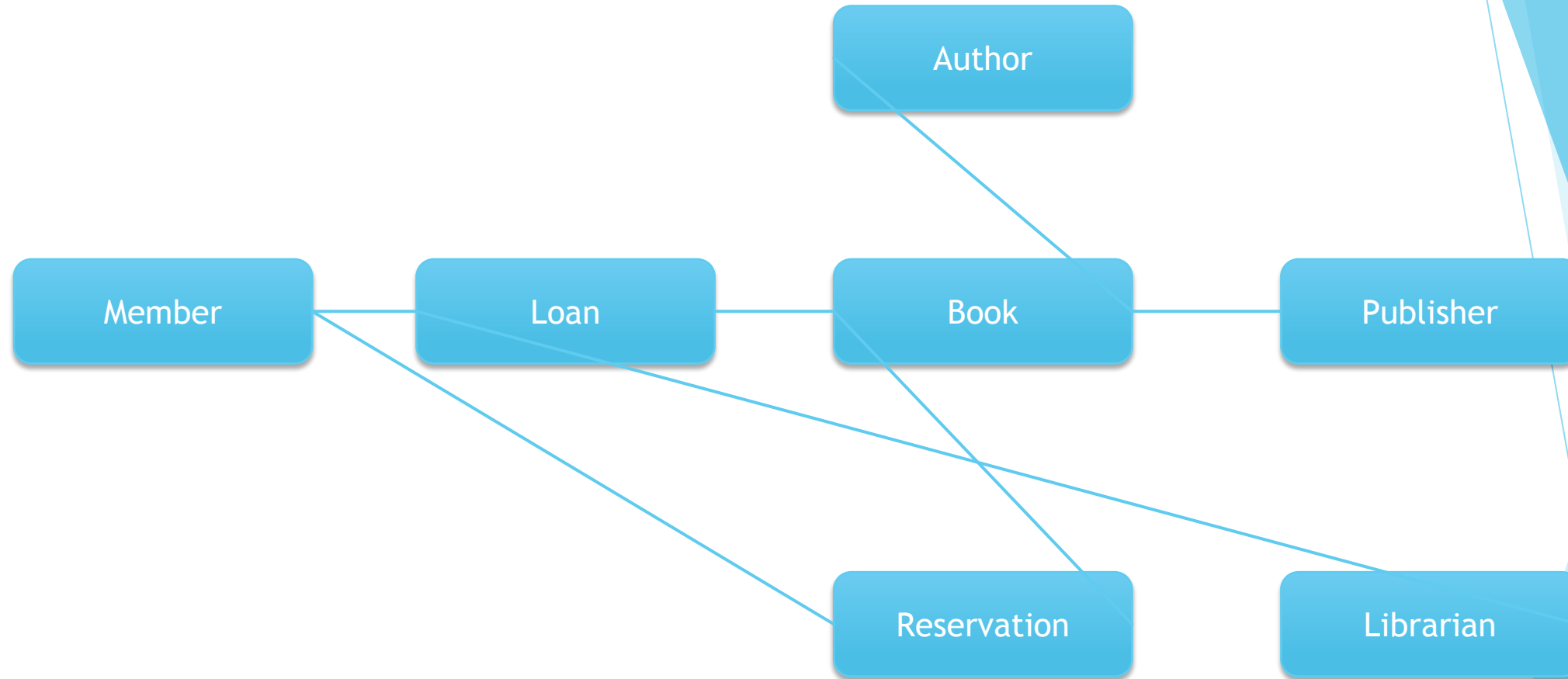
Publisher

Loan

Reservation

Librarian

ER Diagram



Entity Description

Member

Stores member details. PK: MemberID

Book

Stores book information. PK: BookID

Author

One author writes many books.

Publisher

Publishes many books.

Loan

Issue/Return records. FK:
MemberID, BookID

Reservation

Tracks reserved books.

Librarian

Manages library operations.

Advantages of ER Diagram

- Easy database design
- Reduces data redundancy
- Maintains data integrity
- Improves relationship visualization
- Supports database normalization

Relationships

- One Member can borrow many Books.
- One Book can have multiple Loan records.
- One Author can write many Books.
- One Publisher publishes many Books.
- Members can reserve unavailable Books.

Conclusion

- ER Diagram provides a clear database structure.
- It identifies entities, attributes and relationships.
- Helps developers create efficient library databases.
- Useful for real-world Library Management Systems.